



NCEAC . FORM . 001-D

**National Computing Education Accreditation Council**  
**NCEAC**  
**COURSE DESCRIPTION FORM FAST-NUCES**



**INSTITUTION** \_\_\_\_\_

**PROGRAM (S) TO BE EVALUATED**      BSCS  

**A. Course Description**

(Fill out the following table for each course in your computer science curriculum. A filled out form should not be more than 2-3 pages.)

|                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Course Code</b>                                            | CS2006                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Course Title</b>                                           | Operating Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Credit Hours</b>                                           | 3+1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Prerequisites by Course(s) and Topics</b>                  | PF & Data Structures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Assessment Instruments with Weights</b>                    | <ul style="list-style-type: none"><li>• Semester Assessments – 20% ( 2 Prog. Assignments 4 + 3 Quizzes 6 + Project 10 )</li><li>• Midterms – 30%</li><li>• Final Exam – 50%</li></ul> <div style="border: 1px solid black; padding: 5px; margin-top: 10px;"><ul style="list-style-type: none"><li>- <b>Late policy:</b> 50% detection after 48 hours. No awards whatsoever after ONE WEEK after the deadline.</li><li>- <b>Plagiarism</b> punishment up to 20 weightage.</li></ul></div> <p>❖ Project scope limited to <u>Multithreaded Multiprocesses Interprocess communication programming ONLY</u>. Proposals based on theoretical aspects and Socket programming shall not entertained.</p> <p>Marks distribution: 5% proposal , 70 Coding (complexity + proposed outcome), 25% viva &amp; presentation</p>              |
| <b>Course Coordinator</b>                                     | Dr. Nadeem Kafi Khan                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Grading Policy</b>                                         | <b>Absolute grading. Student will receive grades based on predetermined cutoff levels.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Current Catalog Description</b>                            | This course aims to equip students with a solid understanding of operating systems, covering key areas such as system basics, process management, threads and concurrency, scheduling, concurrency and inter-process communication, memory allocation, and security measures. It also emphasizes practical skills like system programming and debugging and virtualization and containers in the context of operating systems. Students will gain hands-on experience through class assessments focusing on system design principles. The goal of this course is to provide students with a comprehensive understanding of operating systems, enabling them to develop efficient applications and address challenges in multi-user, multitasking, and distributed computing environments, while emphasizing practical skills. |
| <b>Textbook (or Laboratory Manual for Laboratory Courses)</b> | <b>Operating system Concepts by Silberchatz, 10th Edition</b><br>(Please do not use Global Edition)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Reference Material</b>                                     | <ul style="list-style-type: none"><li>• OPERATING SYSTEMS INTERNALS, 9th Ed. by Dr. William Stallings</li><li>• Modern Operating System by Andrew S. Tannenbaum 5th Edition.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

|                     |                                                                                                                                                               |                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                       |                       |              |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|--------------|
| <b>Course Goals</b> | <b>A. Course Learning Outcomes (CLOs)</b>                                                                                                                     |                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                       |                       |              |
|                     | <b>CLO</b>                                                                                                                                                    | <b>Name</b>                                                                                                                                                                                                                                                                                           | <b>Domain</b>                                                                                                                                                                                         | <b>Taxonomy Level</b> | <b>Tools</b> |
|                     | 01                                                                                                                                                            | <b>Understand / Describe / discuss / Comprehend</b> <ul style="list-style-type: none"> <li>- Services provided by the operating systems</li> <li>- Virtualization</li> <li>- Concurrency</li> <li>- Persistence</li> <li>- Security</li> </ul>                                                        | Cognitive                                                                                                                                                                                             | 2                     | A,M,F        |
|                     | 02                                                                                                                                                            | <b>Analyze, Compare, Contrast, and evaluate</b> <ul style="list-style-type: none"> <li>- Mechanism of scheduling task</li> <li>- Implementation of concurrency and synchronization mechanism</li> <li>- Performance issues</li> </ul>                                                                 | Cognitive                                                                                                                                                                                             | 3                     | A,M,F        |
|                     | 03                                                                                                                                                            | <b>Design and Implement programs using processes and threads. For example,</b> Simple Operating System Shell, File System Implementation, Process Scheduling Simulator, Interprocess Communication Mechanisms, System Calls Extension, Security Features Implementation, Performance Monitoring Tool. | Cognitive                                                                                                                                                                                             | 3,4                   | A,M,F,P      |
|                     | Tool: A = Assignment, M = Midterm, F=Final, P = Project                                                                                                       |                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                       |                       |              |
|                     | <b>B. Program Learning Outcomes</b>                                                                                                                           |                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                       |                       |              |
|                     | For each attribute below, indicate whether this attribute is covered in this course or not. Leave the cell blank if the enablement is little or non-existent. |                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                       |                       |              |
|                     | <b>PLO 1</b>                                                                                                                                                  | Computing Knowledge                                                                                                                                                                                                                                                                                   | Apply knowledge of mathematics, natural sciences, computing fundamentals, and a computing specialization to the solution of complex computing problems.                                               |                       | ✓            |
|                     | <b>PLO 2</b>                                                                                                                                                  | Problem Analysis                                                                                                                                                                                                                                                                                      | Identify, formulate, research literature, and analyze complex computing problems, reaching substantiated conclusions using first principles of mathematics, natural sciences, and computing sciences. |                       | ✓            |
| <b>PLO 3</b>        | Design/Develop Solutions                                                                                                                                      | Design solutions for complex computing problems and design systems, components, and processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.                                                            |                                                                                                                                                                                                       | ✓                     |              |
| <b>PLO 4</b>        | Investigation & Experimentation                                                                                                                               | Conduct investigation of complex computing problems using research-based knowledge and research-based methods                                                                                                                                                                                         |                                                                                                                                                                                                       |                       |              |

|               |                                |                                                                                                                                                                    |   |
|---------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>PLO 5</b>  | Modern Tool Usage              | Create, select, and apply appropriate techniques, resources and modern computing tools, including prediction and modelling for complex computing problems.         | ✓ |
| <b>PLO 6</b>  | Society Responsibility         | Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal, and cultural issues relevant to context of complex computing problems. |   |
| <b>PLO 7</b>  | Environment and Sustainability | Understand and evaluate sustainability and impact of professional computing work in the solution of complex computing problems                                     |   |
| <b>PLO 8</b>  | Ethics                         | Apply ethical principles and commit to professional ethics and responsibilities and norms of computing practice.                                                   | ✓ |
| <b>PLO 9</b>  | Individual and Team Work       | Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.                                              | ✓ |
| <b>PLO 10</b> | Communication                  | Communicate effectively on complex computing activities with the computing community and with society at large.                                                    | ✓ |
| <b>PLO 11</b> | Project Mgmt and Finance       | Demonstrate knowledge and understanding of management principles and economic decision making and apply these to one's own work as a member or a team.             |   |
| <b>PLO 12</b> | Life Long Learning             | Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological changes.     |   |

**C. Relation between CLOs and PLOs**

(CLO: Course Learning Outcome, PLOs: Program Learning Outcomes)

|             |   | PLOs |   |   |   |   |   |   |   |   |    | 11 | 12 |
|-------------|---|------|---|---|---|---|---|---|---|---|----|----|----|
|             |   | 1    | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |    |    |
| <b>CLOs</b> | 1 | ✓    |   |   |   |   |   |   |   |   |    |    |    |
|             | 2 |      | ✓ |   |   |   |   |   |   |   |    |    |    |
|             | 3 |      |   | ✓ |   |   |   |   |   |   |    |    |    |
|             | 4 |      |   |   |   | ✓ |   |   |   |   |    |    |    |

|                                                                                                                                   |                                                                                                                                                                  |                                                                         |                |                      |            |
|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|----------------|----------------------|------------|
| <b>Topics Covered in the Course, with Number of Lectures on Each Topic</b><br>(assume 15-week instruction and one- hour lectures) | <b>1. Topics to be covered Theory classes:</b>                                                                                                                   |                                                                         |                |                      |            |
|                                                                                                                                   | <b>Weeks</b>                                                                                                                                                     | <b>List of Topics</b>                                                   | <b>Chap. #</b> | <b>Contact Hours</b> | <b>CLO</b> |
|                                                                                                                                   | 1                                                                                                                                                                | Introduction and OS basics (22 <sup>nd</sup> Jan)                       | 1              | 0.5                  | 1          |
|                                                                                                                                   | 2                                                                                                                                                                | Introduction to Operating system                                        | 1              | 5.5                  | 1          |
|                                                                                                                                   | 3                                                                                                                                                                | Operating system structure                                              | 2              | 3                    | 1          |
|                                                                                                                                   | 4                                                                                                                                                                | Processes                                                               | 3              | 3                    | 1,2,3      |
|                                                                                                                                   | 5                                                                                                                                                                | Process Scheduling Algorithm                                            | 5              | 3                    | 1,2,3      |
|                                                                                                                                   | 6                                                                                                                                                                | <b>Mid Term 1 (26<sup>th</sup> Feb, 2024)</b>                           |                |                      |            |
|                                                                                                                                   | 7                                                                                                                                                                | Threads and Concurrency                                                 | 4              | 3                    | 1,2,3      |
|                                                                                                                                   | 8                                                                                                                                                                | Process Synchronization Tools and Examples                              | 6,7            | 3                    | 2,3        |
|                                                                                                                                   | 9                                                                                                                                                                | Deadlocks                                                               | 8              | 3                    | 2,3        |
|                                                                                                                                   | 10                                                                                                                                                               | Main Memory                                                             | 9              | 3                    | 1,2        |
|                                                                                                                                   | 11                                                                                                                                                               | Virtual Memory                                                          | 10             | 3                    | 1,2        |
|                                                                                                                                   | 12                                                                                                                                                               | <b>Mid Term 2 (8<sup>th</sup> Apr, 2024)</b>                            |                |                      |            |
|                                                                                                                                   | 13                                                                                                                                                               | I/O Systems, File-System Interface                                      | 12             | 3                    | 1,2        |
|                                                                                                                                   | 14                                                                                                                                                               | Virtual Machines                                                        | 18             | 3                    | 1,2        |
| <b>Laboratory Projects / Experiments Done in the Course</b>                                                                       | 15                                                                                                                                                               | Revision/completion of topics - 1                                       | 1              | 3                    | 1,2        |
|                                                                                                                                   | 16                                                                                                                                                               | Revision/completion of topics - 2                                       |                |                      |            |
|                                                                                                                                   | 17                                                                                                                                                               | Project Presentations and Viva (6 <sup>th</sup> - 10 <sup>th</sup> May) |                |                      |            |
|                                                                                                                                   | <b>2. Topics to be covered in Labs:</b>                                                                                                                          |                                                                         |                |                      |            |
|                                                                                                                                   | Lab 1: Introduction & Basic Linux Commands and Virtual Box installation                                                                                          |                                                                         |                |                      |            |
|                                                                                                                                   | Lab 2: Creating, Compiling and executing C/C++ programs using gcc/g++ compilers using makefile                                                                   |                                                                         |                |                      |            |
|                                                                                                                                   | Lab 3: Linux Shell Scripting (installations and configurations, system admin, task orchestration)                                                                |                                                                         |                |                      |            |
|                                                                                                                                   | Lab 4: System Call related to Process Management, argument arrays                                                                                                |                                                                         |                |                      |            |
|                                                                                                                                   | Lab 5: POSIX thread programming without synchronization                                                                                                          |                                                                         |                |                      |            |
|                                                                                                                                   | Lab 6: Inter- Process Communication (IPC, Named Pipes and demo basic sockets programming)                                                                        |                                                                         |                |                      |            |
|                                                                                                                                   | Lab 7: Shared Memory and Memory Mapped Files                                                                                                                     |                                                                         |                |                      |            |
|                                                                                                                                   | Lab 8: Mid Exam                                                                                                                                                  |                                                                         |                |                      |            |
|                                                                                                                                   | Lab 9: Multithread Programming with synchronization primitives – 1                                                                                               |                                                                         |                |                      |            |
|                                                                                                                                   | Lab 10: Multithread Programming with synchronization primitives – 2 (using File-System calls)                                                                    |                                                                         |                |                      |            |
|                                                                                                                                   | Lab 11: The Readers and Writers Problem Lab                                                                                                                      |                                                                         |                |                      |            |
|                                                                                                                                   | Lab 12: OS security lab (desktop and server threats, hardening Linux OS, network security basics)                                                                |                                                                         |                |                      |            |
|                                                                                                                                   | Lab 13: System Configuration. Boot loader, Managing Services, System Startup Files (rc.d, rc.sysinit rc.local init.d), make, configure install, Integrity Checks |                                                                         |                |                      |            |
|                                                                                                                                   | Lab 14: Creating a module in Kernel - 1                                                                                                                          |                                                                         |                |                      |            |
|                                                                                                                                   | Lab 15: Creating a module in Kernel - 2                                                                                                                          |                                                                         |                |                      |            |
|                                                                                                                                   | Lab 16: Final Lab Exam (Lab Midterm syllabus will be 20% and rest 80%)                                                                                           |                                                                         |                |                      |            |
|                                                                                                                                   | Lab 17: Lab Project Evaluation and Lab Final Exam Review.                                                                                                        |                                                                         |                |                      |            |



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|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|------------------------|----------------------------------|
| <b>Programming Assignments Done in the Course</b> | <b>2 in Theory and 13+ in different OS labs</b>                                                                                                                                                                                                                                                                       |                         |                        |                                  |
| <b>Class Time Spent on</b><br>(in credit hours)   | <b>Theory</b>                                                                                                                                                                                                                                                                                                         | <b>Problem Analysis</b> | <b>Solution Design</b> | <b>Social and Ethical Issues</b> |
|                                                   | 20                                                                                                                                                                                                                                                                                                                    | 15                      | 6                      | 1                                |
| <b>Oral and Written Communications</b>            | Every student is required to submit at least __1__ written report of typically __2__ pages and to make __1__ oral presentations of typically __10__ minute's duration. Include only material that is graded for grammar, spelling, style, and so forth, as well as for technical content, completeness, and accuracy. |                         |                        |                                  |

**Instructor Name: Dr. Nadeem Kafi Khan**

**Instructor Signature:** \_\_\_\_\_

**Date: 19<sup>th</sup> January, 2024**